



Start Planning

Set some goals for your team and each person on the team. Decide how you will make decisions. What tools will you use to keep organized? How will you make sure everyone is involved?

Sample Resources

[Google Drive](#)
[Google Calendar](#)
[Slack](#)
[Trello](#)
[Tools for Teams](#)

Develop a Team Identity

Pick a team name
 Design a team logo
 Design a team shirt

[Decision-Making Models](#)
[Reaching a Consensus](#)
[Selecting a Team Name](#)
[Developing a Team Identity](#)

Learn the Core Values

What are the FIRST Core Values?
 How can we use them this season?
 What can we learn from FIRST?

[FIRST Vision and Mission](#)
[More Than Robots](#)
[Gracious Professionalism](#)

Learn to Work Together

Learn to collaborate, come to a consensus, and achieve a goal. Select some fun team building activities to help you.

[Core Values Activities](#)
[UNICEF Team Building Activities](#)
[Teaching Expertise - Team Building](#)

Discover Interests and Goals

Find out what students on the team enjoy and what they want to work on/learn

FLL Chicken Challenge

FLLChicken is a fun-loving character who appears in many seasons, including BIOGLOW.

[FLLChicken Facebook Page](#)
[FLLChicken Challenge](#)

Learn the Core Values and Learn to Work Together as a Team



Team Building Activity: What do we have in Common?

Objectives: Through this activity teams will learn more about each other and appreciate one another.

Materials:

- Strips of paper (or post-it notes)
- Poster board to stick them on to with the names of each of the students on the top.
- Set of questions
- A small box to collect responses

Setup:

- Each student is given small pieces of paper to write responses on
- Come up with at least 1 question for each student you have on the team. Examples: 1) How many siblings do you have?, 2) If you could travel anywhere, where would you go?, 3) What is your favorite subject in school?, 4) What is your favorite food to eat?, 5) If you could invent something to help you, what would you like to invent?, 6) Do you play a musical instrument (which one)? 7) What is your favorite animal?

Instructions to Team:

Let's find out what we have in common. I will read a question out loud. Each of you should write your answer on a strip of paper/post-it note. When finished, place your response in this box. When everyone is done writing answers, we will pick out one paper at a time and guess who it could be about.

Jessica	Sam	Christian	Saif	Arjan
Tennis		Clarinet		Guitar
Dog	Soccer	Dog	Soccer	History
1 Sister	Guitar	History	1 Sister	1 Brother

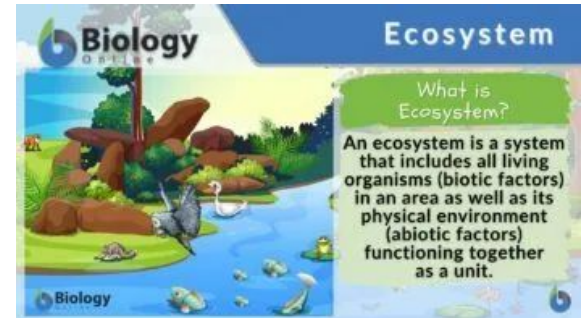


Core Values Activity: Web of Life

Objectives: Discover how all the players in an ecosystem depend on each other to survive and learn what it means to work together as a team.

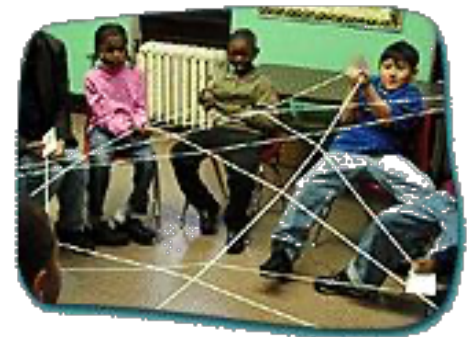
Materials:

- Index cards
- Marker or pen
- Ball of twine/yarn
- List of connections: sun, bumblebee, mushroom, rock, grass, mouse, earthworm, ant, spider, river, woodpecker, rain, dead leaf, butterfly, flower, tree, snake, frog, snail, living leaf, owl, squirrel, soil, deer *[You can modify this list based on the region you live in.]*



Terms to Learn:

- Ecosystem:
- Organism:



Setup:

- Write the names of each organism on a card. Have everyone sit in a circle.

Instructions:

Have students pick up cards so they are distributed equally. The student with the Tree card starts off and throws the yarn to another person. The student who receives the yarn reads their first card out loud and then everyone can discuss how their card interacts with the Tree. The student can then throw the ball of yarn to another student. The recipient of the ball of yarn continues the process, discussing how their card is connected to the previous one. Once everyone all the cards are done, discuss what would happen if one of the organisms was removed from the web.

Discuss how an ecosystem is similar to a FIRST LEGO League team.



Core Value Activity: Scavenger Hunt

Objectives: Discover the natural world around you while working together as a team.

Materials:

- Multiple copies of the graphic on the next page
- Pens/pencils
- If you have younger students, you can create a simplified scavenger hunt or one more specific to your region

Terms to Learn:

- Abiotic:
- Biotic:
- Water Cycle:
- Renewable Energy:
- Predator:
- Prey:
- Herbivore:
- Omnivore:
- Carnivore:
- Pollinator:
- Camouflage:
- Metamorphosis:
- Mammal:
- Food Chain:

Instructions: Go through the vocabulary words together with your coach before the scavenger hunt. Work as a team to find everything in the graphic on the next page. Whenever you complete a space, cross it out. Remember to leave the objects/organisms in their habitat undisturbed. Optional: make a Food Chain using some of the organisms you found.



BIODIVERSITY

scavenger hunt

Record your answers on a separate sheet of paper.

5 nonliving components in an ecosystem! ABIOTIC factors	5 living components in an ecosystem! BIOTIC factors	1 Stage of the WATER CYCLE	2 sources of RENEWABLE ENERGY	1 PREDATOR What does it eat?	1 PREY What eats it?
1 HERBIVORE	1 OMNIVORE	1 CARNIVORE	3 organisms that FLY or SWIM	2 organisms that WALK or CRAWL	1 organism that HOPS
1 POLLINATOR	BREATHE in clean AIR from PLANTS What do plants provide us with?	1 TREE ANIMAL HOME	1 GROUND ANIMAL HOME	1 WATER ANIMAL HOME	LIFT UP 1 weathered ROCK How many bugs do you see?
How many different SEEDS can you find?	EXAMINE SOIL What's in it?	1 organism that is CAMOUFLAGED	1 organism that goes through METAMORPHOSIS	Find something a BIRD would eat	Find something a MAMMAL would eat
Based on your observations , create a food chain using some of the organisms you spotted in this ecosystem !					





1

Robot Prototyping

Look at the layout of the mat and where the models are. What type of robot would be useful? How wide can the robot be? Are there lines to follow? What sensors would help with navigation (e.g. color/gyro)? Where should the sensors go on the robot? What type of wheels should you use?

Use the pre-season to build different robots and test them. It is okay to start with a basic design that you modify to suit your goals. Document your tests and results. FLLTutorials Robot Design Worksheets has examples of how to document these tests.

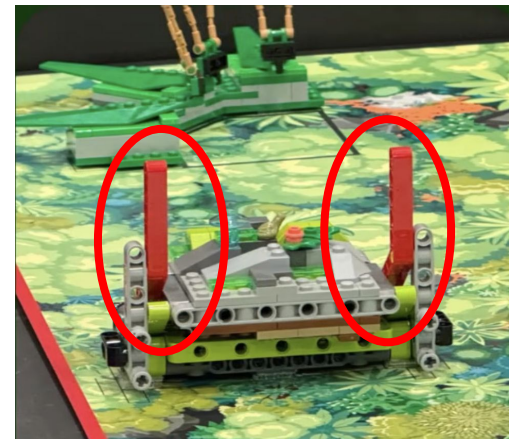
Robot Testing		Name:	
Instructions: 1. If you design more than one robot, use this chart to compare them. At the top of each column, describe your robot. 2. Come up with some basic tests to compare the robot designs. Can this robot move straight accurately? Can it turn consistently? Can it line follow? Can it detect a line? Did the robot move as intended? 3. Discuss which robot performed the best to help you pick the best design for your team.			
	Robot 1:	Robot 2:	Robot 3:
	Wheels:	Wheels:	Wheels:
	Size:	Size:	Size:
	Sensors:	Sensors:	Sensors:
	Motors:	Motors:	Motors:
Move Straight 50cm			
Overall: Speed Balance			

- [EV3 Base Robot Designs](#)
- [Spike Prime Base Robot Designs](#)
- [Additional Robot Designs](#)
- [Lessons about Wheels/Sensor Placement](#)

Mechanism Prototyping

Generally, *FIRST* LEGO League missions require similar actions -e.g. pull, push, lift, turn, or drop off. The Future Edition also shows collection and tossing of round balls. Take a close look at the BIOGLOW [teaser video](#) and watch for how the models are activated. The teaser actually reveals many of the actions that will be needed. Also look at images released. e.g. What do you think the red liftarms/connectors on the different mission models are for?

Learning to build simple mechanisms that might achieve these tasks can help you prepare for any season.



How do these and other models move in the [teaser video](#)? What gets released or open? What objects might you have to move? Are there objects with loops?



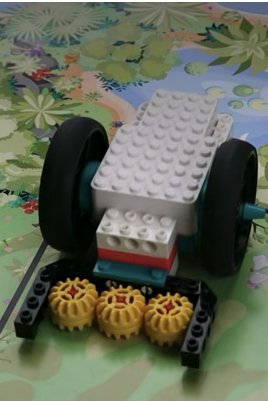
- [Books by Yoshihito Isogawa](#)
- [Technic Builder's Guide](#)
- [Pre-Season Building Challenges](#)

1

Robot and Mechanism Prototyping

2

Look carefully at the [Future Edition Showcase](#) video. What types of robots/mechanisms are needed for each role and how can you prepare? Although the actual rules are not released yet, the video reveals some of the actions that will be needed and the size of the objects that will need to be moved or delivered. Use this information to practice building skills.



Driver: Uses the robot and controller to navigate the field. Robot and attachments deliver resources and keystone species to the rest of the gamefield

→ Practice designing different drive bases with your CS&AI set and programming the controller. Build some attachments to collect and move both resources and keystone species.



Operator: Uses a mechanical tool to supply resources to the rest of the game field

→ Practice building tools that launch resources.



Technician: Uses a robotic tool to transport keystone species and trigger actions

→ Practice building a mechanism that can launch keystone species.



Resources and Keystone Species

Try building some of these mechanisms. You will have to adapt the designs to use parts you have or modify them to fit the size/shape of the resources or keystone species. Think about how you could add motors or sensors from the CS&AI kit to create a robotics tool.

- LEGO Catapult: [Frugal Fun](#), [Knorr LEGO](#), [LEGO Edu Simple Machines](#), [Technic Catapult](#)
- LEGO Trebuchet: [Brick Science](#),
- LEGO Kickers: [BricQ Motion Free Kick](#)
- LEGO GBC: [NXT](#), [Akiyuki](#), [LEGO GBC](#), [Dual Liftarm GBC](#), [Catapult GBC](#), [Catapult by Dunes](#),
- Robot Candy Launchers - [JKBrickworks](#)



3

Programming Skills

Look at the Future Edition set and Showcase Video. What skill might be worth learning?

You can see the color sensor being used to activate a tool, you can see the controller being used to move a robot as well as a tool. You also see the Grand Tree turning. Without any additional details, the following skills can be useful:

- Programming the controller
- Using a color sensor to activate a motor/launcher
- Controlling actions with the double motor

Programming Skills

Look at the Founders Edition set. What skills might be worth learning? Discuss how some of these techniques can help you navigate the mat reliably. Some techniques may be more useful than others.

- Moving Straight and turning - basic movements needed for any navigation
- Moving Until a Line - allows you to know exactly where you are on a mat
- Squaring on a Line - allows you to straighten out to a perpendicular line
- Wall Following - allows you to be straight along a wall
- Backing into a Wall - allows you to straighten out if you are crooked